Module -3 Question Bank

MCQS

1) In Vedanga _______can be considered as a user manual that provides instructions and directions to lead all aspect of life.

a. Siksa
b. Vyakrana
c. Jyotish
d. Kalpa

2) The knowledge of the movement of stars and planetary bodies is called ______in Vedanga
a. Siksa
b. Vyakrana
c. Jyotish
d. Kalpa

3) _______was the founder of Visistadvaita Vedanta
a. Shankaracharya
b. Madhavacarya
c. Ramanuja
d. Jaimini

4) _______was the founder of Dvaita Vedanta
a. Shankaracharya
b. Madhavacarya
c. Ramanuja
d. Jaimini

5) _________________gave the THEORY OF binary numeral system, arithmetical triangle.
a. Baudhayan
b. Pingala
c. Shalya
d. None of the Above

6) Who among the following in NOT an ancient Indian Mathematicians
a) Vyasa
b) Pingala
c)Aryabatta
d) Bhramagupta

7) Which Indian mathematician believed to invented zero?
a) Bhaskara II
b)Aryabatta
c)Pingala
d)Bhramagupta

8) Who is notable for being the last of the Vedic mathematicians, who wrote the Katyayana Sulba Sutra, which presented much geometry, including the general Pythagorean theorem and a computation of the square root of 2 correct to five decimal places.
a) Aryabatta
b) Bhaskara II
c) Katyayana
d) None of these

9) ....Mathematicians were apparently also the first to use the words hunya (literally void in Sanskrit) to refer to zero.
a) **Jain**
b) Buddhist
c) Hindu
d) Mohmmedan.

10) According to Hayashi,…………….contain "the earliest extant verbal expression of the Pythagorean Theorem in the world, although it had already been known to the Old Babylonians."

a)Katayana
b) **Sulba Sutra**
c) Both a & b
d) None of these

11) Who discovered the right triangle?

a) Pythagoras
b) Ancient Egyptians
c) Miley Cyrus
d) Michelangelo

12) Who invented numbers?

a) Brahmagupta
b) Archimedes
c) Euclid
d) Fibonacci

13) Which branch of mathematics deals with shapes and sizes?
a) Algebra
b) Geometry
c) Calculus
d) Arithmetic

14) Which ancient Indian text contains early references to the concept of zero?
a) Ramayana
b) Sulbasutras
c) Aryabhatiya
d) Brahmasphutasiddhanta

15) Who is the author of the famous book Arthashastra, which also includes mathematical concepts?

a) Aryabhata
b) Bhaskaracharya
c) Chanakya
d) Brahmagupta

16) What is the study of numbers and their properties called?

a) Geometry
b) Algebra
c) Number Theory
d) Calculus

17) Which mathematician improved the sine tables created by Aryabhata?

a) Varahamihira
b) Aryabatta
c) Bhramagupta
d) all the above

18) Who is credited with proposing a heliocentric model of the solar system that predates Copernicus?

a) Aryabhata I
b) Bhaskara II
c) Nilakantha somayaji
d) Varahamihira

19) Which astronomical system did Varahamihira consider the best in his book "Panchasiddhantika" ?

a) Aryasidhanta
b) Suryasidanta
c) Vasistasidanta
d) Romakasiddhanta

20) In the context of Indian astronomy, what does the term "Ahargana" refer to?

a) The number of solar months in a year
b) The number of lunar phases in a year
c) The number of constellations in the zodiac
d) The number of civil days from specific epoch

21) Which correction is applicable to the non-circular nature of planetary orbits in Siddhantic astronomy?

a) Sighra correction
b) **Manda correction**
c) Bija Samskara correction
d) Parallax correction

22) Which of the following celestial bodies is NOT mentioned as one of the taragrahas in the text?
a) Mercury
b) Venus
c) Mars
d) Neptune

23) What are the Vedangas?
a) Six ancient astronomical text
b) Six branches of learning linked to vedas
c) Six types of celestial bodies
d) None of these

24) Who among the following developed the star positioning instrument in ancient India?

a) Lalla
b) Ganesh Daywanya
c) Shripati
d) Bhaskaracharya

25) **Consider the following.**

I. Siddhanta Siromani is the major treatise of Indian mathematician Bhāskaracharya.
II. Bhāskaracharya wrote the Siddhanta Siromani in 1150 when he was 36 years old.
Which of the following is/are correct about Bhaskaracharya?
a) Only I
b) Only II
c) Both I and II
d) Neither I nor II

Descriptive Questions

1. Write about the contribution of Arybhata in Astronomy
2. List and describe any 5 great mathematicians and their contributions
3. State arithmetic and explain Indian Vedic geometry
4. Explain Binary mathematics and combinatorial problems in Chandah-śāstra of Pingala
5. State and explain Celestial Coordinate System.
6. What are the advantages of the Indian numeral system over the Roman numerals? Explain with the help of examples?
7. Explain when was the concept zero discovered in India? What is its special significance?
8. Illustrate the difference between the katapayadi system and Bhuta samkya system?
9. Briefly explain about the knowledge of π that Indian mathematicians possessed. How does it compare with that of other mathematicians?
10. Explain the solar and lunar month in an Indian calendar. Explain the term ‘adhikamasa’.
11. Explain the notion of the ecliptic and the northern and southern motions of the sun?
12. Briefly describe the instruments used by ancient Indians for astronomical purpose?